

Japman Singh

+91 79862-12038 | japman006@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EXPERIENCE

Summer Research Intern, Annam.ai

June – July 2026

Wind-Sensing Research Project

IIT Ropar

- Developed and debugged embedded C firmware for ultrasonic transducers on TI MSP430 microcontrollers to measure Time-of-Flight (ToF) and estimate wind speed and direction
- Implemented signal averaging and filtering to improve sensor measurement stability, and transmitted processed data via UART for real-time analysis

Software Intern, iHub – AWaDH @ IIT Ropar

June – July 2025

Project: Soil Spectra

IIT Ropar

- Preprocessed a 19,000+ sample hyperspectral soil reflectance dataset (1,000 features, 400–940 nm) using Python, Pandas, and NumPy: duplicate removal, outlier treatment, missing-value handling, and scaling
- Optimized interpolation step size and data cleaning pipelines for soil nutrient prediction models, achieving R^2 of 0.803 (Organic Carbon) and 0.690 (Nitrogen) using scikit-learn
- Evaluated multiple regression models to predict soil nutrients (Organic Carbon, Nitrogen, Phosphorus, Potassium) from spectral reflectance data, analyzing RMSE and R^2 across nutrients to identify best-fit configurations
- Collaborated within a cross-functional team (Android + AI/ML) building an end-to-end IoT-based precision agriculture system with BLE-connected optical soil sensors

PROJECTS

Rural Financial Advisor

[GitHub](#) | [Live Demo](#)

AI-Assisted Loan Advisor for Rural Entrepreneurs

Java, Spring Boot, PostgreSQL, Docker, AWS EC2

- Led a 3-member team at the WeMakeDevs × AWS First Commit hackathon; built the Spring Boot (Java 17) and PostgreSQL backend for loan sizing, EMI (repayment capped at 50% of income), PM Mudra/PMEGP scheme matching, and credit-risk bands via REST APIs
- Made the backend the single source of truth for financial figures so the AI service (Amazon Bedrock) only explains validated numbers, with a rule-based fallback if the model is unavailable
- Integrated React, Spring Boot, and a FastAPI AI service behind nginx with Docker Compose; deployed on AWS EC2 using an IAM role (no stored credentials) with only port 80 public

Online Chess

[GitHub](#) | [Live Demo](#)

Chess Rules Engine & Two-Player Web App

Java, Spring Boot, React, JUnit, AWS EC2, nginx

- Built a chess rules engine in Java from scratch: moves that leave the player's own king in check are rejected, giving check, checkmate, stalemate, and promotion; verified with JUnit tests on known positions
- Designed a Spring Boot REST API (5 endpoints) for real-time two-device play with invite codes, per-player tokens, server-side turn enforcement, and per-game locking; React UI polls state every second
- Deployed on AWS EC2 with an nginx reverse proxy, a systemd-managed Spring Boot service, and a restricted security group

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, HTML, CSS

Frameworks & Libraries: Spring Boot, React, Tailwind CSS

Databases & Cloud: PostgreSQL, AWS (EC2, IAM), nginx, Docker, Docker compose

Developer Tools: Git, IntelliJ IDEA, VS Code, Maven, JUnit, Postman

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming, RESTful APIs, MVC Architecture

Achievements: 450+ DSA problems solved; [LeetCode](#) rating: 1700+; [Codeforces](#) rating: 1200+

EDUCATION

Punjabi University

B.Tech in Computer Science

Patiala, Punjab, India

Aug. 2023 – Aug. 2027